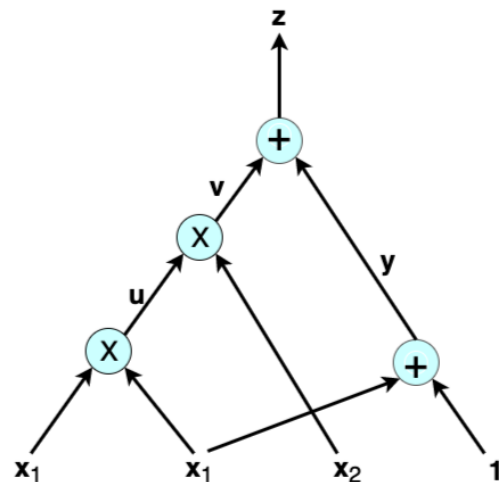


Intelligently Detecting Underconstrained ZK Pipelines

Yavor Litchev

Introduction

Zero-knowledge (ZK) applications encode program correctness as arithmetic circuits compiled into Rank-1 Constraint Systems (R1CS). Under-constrained (UC) circuits are a recurring failure mode: missing relations admit invalid witnesses that still satisfy the published constraints.



Example R1CS Circuit

- This project develops a lightweight, front-end-agnostic detector that ingests R1CS JSON from Circom, Gnark, and a Noir subset and returns a calibrated probability that a circuit is UC
- The detector combines structural statistics of the constraint matrices with probe-time Jacobian rank features from randomized algebraic checks
- The resulting score is designed to prioritize high-risk circuits for subsequent formal verification or fuzzing rather than replace those methods

Data

Real labeled data for UC circuits is scarce, so we construct a seeded corpus from 4–6 canonical gadget families (Merkle path, range check, Poseidon-round chain, public-input linkage, and tiny_add). For each base circuit and random seed we generate a matched pair:

- A semantics-breaking UC variant that preserves syntactic validity but relaxes constraints (e.g., dropping rows, linearizing multiplicative gates, omitting Poseidon links, or removing range bounds)
- A semantics-preserving control (e.g., row permutations)

This pairing yields balanced labeled subsets while additional unlabeled circuits from other front-ends are available to the unsupervised core.

"Canonical gadgets" → "UC mutation" and "Control edit" → "Labeled pairs"

Field	Value / Notes
Canonical gadget families	5: Merkle, Range check, Poseidon, Public-input linkage, tiny_add
Labeling scheme	Paired per base seed → (i) UC variant vs (ii) Control (semantics-preserving)
UC mutations (examples)	Drop a small % of rows, zero selected variable columns, etc.
Control edits (examples)	Row permutations, benign row duplicates, structurally neutral rewrites
Front-ends	Circom/snarkjs (seedbank), Gnark & minimal Noir adapters
Split protocol	Group-aware by 'parent_id' (all siblings of a base circuit go to train or test)
Primary goal	High recall at low FP (1%/5%) for triage, easy laptop-scale reproducibility

Features

From the sparse R1CS matrices A, B, C with m constraints and n variables, we extract:

- Structural features, such as the constraint–variable ratio m/n , average fan-in and fan-out, etc.
- Algebraic behavior, via randomized Jacobian probing over $\mathbb{F}_p$: We freeze public inputs, perturb witness coordinates, form the Jacobian $J = \partial F / \partial w$ of $F(w) = (Aw) \odot (Bw) - Cw$, and record properties
- We aggregate probe statistics (means and standard deviations across probes) and standardize all features, optionally applying PCA before density modeling. Rank can be computed via finite-field linear algebra or floating-point SVD as a control, but the deployed configuration uses finite-field rank to respect ZK arithmetic

Feature group	Examples / Notes
Structural (R1CS)	$ratio_{cv} = m/n$, average fan-in, average fan-out, mult_share, deg_proxy, counts
Probe-time algebraic	$\mathbb{F}_p \rightarrow J = \partial F / \partial w$, record rank_mean/std, nullity_mean/std, etc.
Probe config	Default trials = 10, subset = 32, rank mode = 'algebraic' (finite-field)
Preprocessing	Standardize all numeric features, optional PCA before density modeling
Targets / outputs	$label_{UC} \in \{0, 1\}$ for calibration, detector outputs prob_buggy and LLR

Models

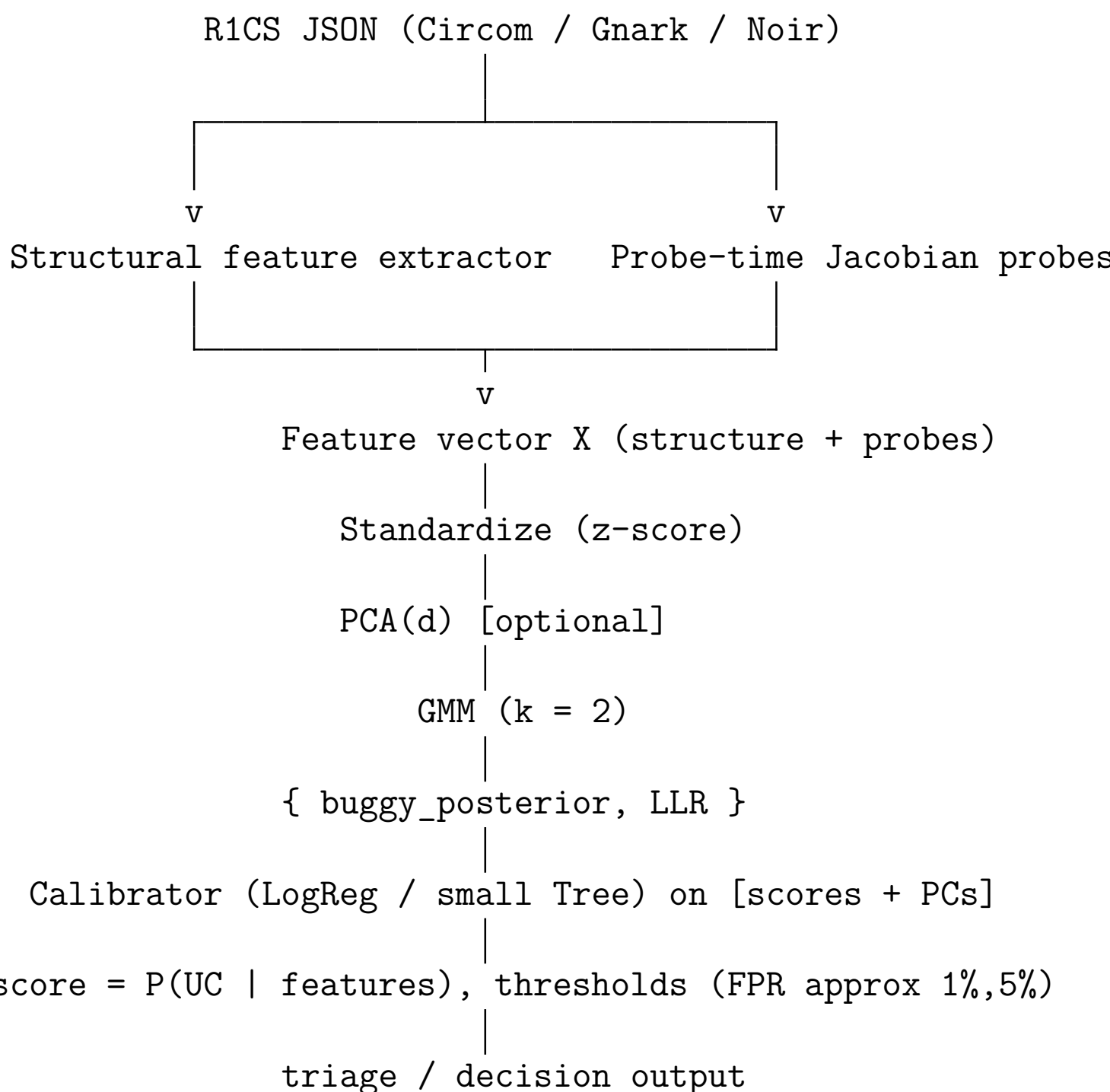
Let $x_i \in \mathbb{R}^d$ denote the standardized feature vector for circuit i . Optionally, we project to the first r principal components $u_i = P_r^\top x_i$. A two-component Gaussian Mixture Model (GMM) is fit via EM,

$$p(u) = \sum_{k=1}^2 \pi_k \mathcal{N}(u \mid \mu_k, \Sigma_k), \quad \sum_k \pi_k = 1, \pi_k \geq 0,$$

and the component whose posterior mass aligns with seeded UC variants is labeled “buggy.” For each circuit we compute the posterior membership in the buggy component and a log-likelihood ratio (LLR) against the non-buggy component; these scores, together with a few top PCA coordinates, form a low-dimensional vector z_i that is passed to a thin logistic calibrator

$$\hat{p}_{UC}(i) = \sigma(\beta_0 + \beta^\top z_i),$$

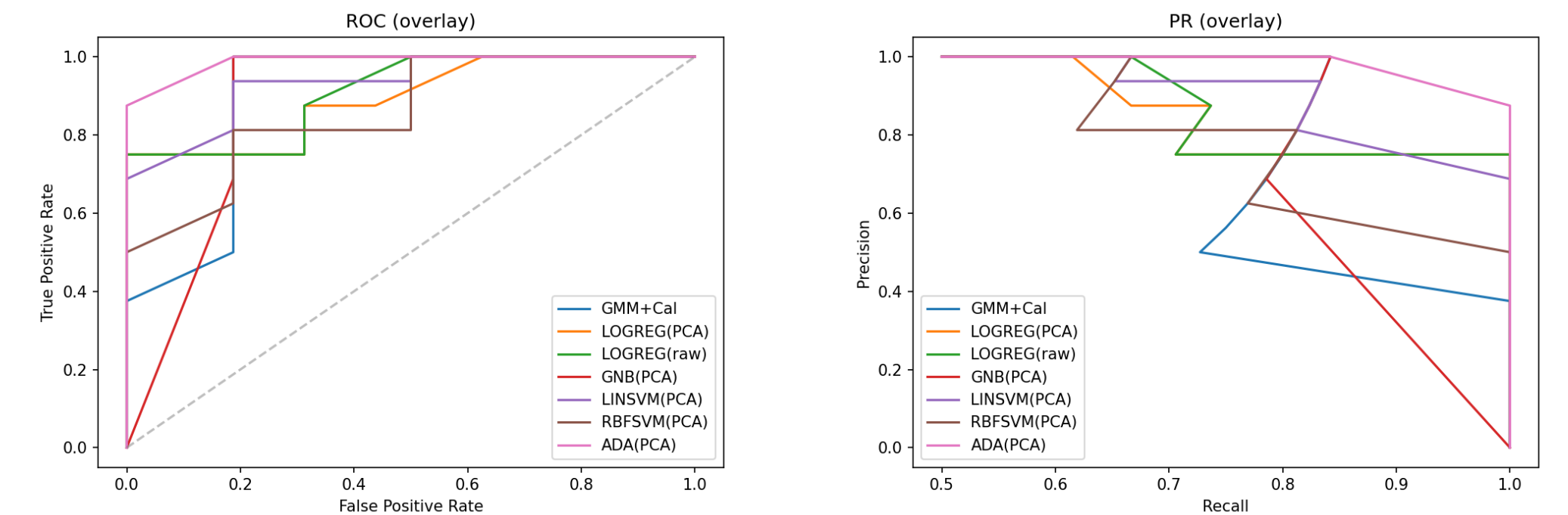
which outputs a calibrated probability of under-constraint and operating thresholds (e.g., FPR $\approx$ 1%, 5%). As supervised baselines, we train logistic regression, Gaussian Naive Bayes, linear and RBF-kernel SVMs, and AdaBoost with depth-1 decision trees on the same standardized features.



Results

Model	AUROC	AP	TPR@1%	TPR@5%	Acc	Prec	Recall	F1	Brier
GMM(PCA)	0.977 ↑	0.969 ↑	0.812 ↑	0.812 ↑	0.875 ↑	0.833 ↑	0.938 ↑	0.882 ↑	0.077 ↓
GMM(PCA)-noProbe	0.895 ↑	0.867 ↑	0.375 ↑	0.375 ↑	0.656 ↑	0.727 ↑	0.500 ↑	0.593 ↑	0.177 ↓
GMM(raw)	0.977 ↑	0.969 ↑	0.812 ↑	0.812 ↑	0.875 ↑	0.833 ↑	0.938 ↑	0.882 ↑	0.077 ↓
GMM(raw)-noProbe	0.895 ↑	0.867 ↑	0.375 ↑	0.375 ↑	0.656 ↑	0.727 ↑	0.500 ↑	0.593 ↑	0.203 ↓
LOGREG(PCA)	0.965 ↑	0.958 ↑	0.750 ↑	0.750 ↑	0.844 ↑	0.824 ↑	0.875 ↑	0.848 ↑	0.119 ↓
LOGREG(PCA)-noProbe	0.895 ↑	0.919 ↑	0.750 ↑	0.750 ↑	0.719 ↑	0.706 ↑	0.750 ↑	0.727 ↑	0.219 ↓
LOGREG(raw)	0.965 ↑	0.958 ↑	0.750 ↑	0.750 ↑	0.844 ↑	0.824 ↑	0.875 ↑	0.848 ↑	0.138 ↓
LOGREG(raw)-noProbe	0.910 ↑	0.925 ↑	0.750 ↑	0.750 ↑	0.750 ↑	0.719 ↑	0.706 ↑	0.750 ↑	0.727 ↑
GNB(PCA)	0.877 ↑	0.798 ↑	0.000 ↑	0.000 ↑	0.750 ↑	0.786 ↑	0.688 ↑	0.733 ↑	0.121 ↓
LINSVM(PCA)	0.934 ↑	0.934 ↑	0.688 ↑	0.688 ↑	0.812 ↑	0.812 ↑	0.812 ↑	0.812 ↑	0.155 ↓
RBFSVM(PCA)	0.859 ↑	0.870 ↑	0.500 ↑	0.500 ↑	0.719 ↑	0.769 ↑	0.625 ↑	0.690 ↑	0.167 ↓
ADA(PCA)	0.988 ↑	0.980 ↑	0.875 ↑	0.875 ↑	0.906 ↑	0.842 ↑	1.000 ↑	0.914 ↑	0.120 ↓

On a held-out, group-aware split by gadget family, the compact detector (standardized structural + probe features, no PCA, GMM with (k=2), logistic calibrator, (T=10) probes) achieves AUROC $\approx$ 0.98, Average Precision $\approx$ 0.97, and TPR@1% FP $\approx$ 0.81, satisfying the low-false-positive triage requirement. Probe-ablation and rank-mode experiments show that removing probe-time algebraic features roughly halves TPR@1% FP, while structural-feature ablations cause only mild degradation, and finite-field and SVD-based ranks are competitive on this corpus.



Rank mode	AUROC	AP	TPR@1%	TPR@5%
mod-p	0.977 ↑	0.969 ↑	0.812 ↑	0.812 ↑
float-SVD	0.988 ↑	0.980 ↑	0.875 ↑	0.875 ↑

Discussion

Probe-time algebraic features are essential for detecting under-constraints at low FPR: removing Jacobian rank, nullity, and wiggle-rate statistics approximately halves TPR@1% FP for otherwise identical models. Structural features alone capture coarse size and topology differences but cannot observe the local “wiggle spaces” introduced by missing constraints, underscoring the need for randomized Jacobian checks.

- The unsupervised-first architecture (GMM capturing the bimodal geometry of healthy versus buggy circuits, followed by a thin supervised calibrator) fits the low-label regime and yields well-behaved probabilities with improved Brier scores
- Cross-family experiments show that performance transfers to unseen gadget families, though metrics degrade modestly on small designs (e.g., tiny_add), where structural and probe diversity are limited
- False negatives arise primarily when UC variants remove very few constraints or alter low-influence rows with weak local sensitivity, while false positives arise from benign circuits with atypical sparsity or clusters of near-duplicate rows, suggesting that richer probes and sparsity-aware features could further sharpen the detector. Overall, the system functions effectively as a triage front-end: it ranks circuits by risk so that expensive formal verification and fuzzing can focus on the most suspicious designs

Future

Future work will expand the seedbank to include larger real-world applications and additional constraint systems (e.g., PLONKish/ACIR), enabling more robust training and external validation. We plan to design richer algebraic probes, such as structured perturbations and higher-order sensitivities, and to integrate the detector more tightly with formal verifiers and fuzzers in a learned triage loop. Finally, we aim to scale feature extraction to very large circuits and develop row- and gate-level attribution tools so that high-risk scores are accompanied by human-interpretable explanations that directly support developer remediation.

References

- Hao Chen, Guoqiang Li, Minyu Chen, Ruibang Liu, and Sinka Gao. AC4: Algebraic computation checker for circuit constraints in zero knowledge proofs, 2024.
- Alessandro Coglio, Eric McCarthy, and Eric W. Smith. Formal verification of zero-knowledge circuits. In Bruno Bernardo and Diego Marmosier, editors, *Proceedings of the 5th Workshop on Formal Methods for Blockchains (FMBC 2023)*, volume 393 of *Electronic Proceedings in Theoretical Computer Science*, pages 112–125, 2023.
- Shankara Pailoor, Yanju Chen, Franklyn Wang, Silva Zhang, Isil Dillig, and Yuepeng Wang. Automated detection of under-constrained circuits in zero-knowledge proofs. *Proceedings of the ACM on Programming Languages*, 7(PLDI), 2023.
- Trail of Bits. Circomspect: A static analyzer and linter for the circom zero-knowledge DSL. <https://github.com/trailofbits/circomspect>, 2024. Accessed: 2024.
- Veridise. Picus: Underconstrained circuit detector — overview. <https://docs.veridise.com>, 2025. Documentation.
- Hongbo Wen, Jon Stephens, Yanju Chen, Kostas Ferles, Shankara Pailoor, Kyle Charbonnet, Isil Dillig, and Yu Feng. Practical security analysis of zero-knowledge proof circuits. In *33rd USENIX Security Symposium (USENIX Security 24)*, Philadelphia, PA, 2024. USENIX Association.